

Food Safety Traceability Technology and Application Based on Blockchain

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Abstract—In the blockchain traceability system, consumers mainly build trust in the system through system quality, and a good operating experience improves consumer efficiency and experience, thereby enhancing trust. In contrast, consumers build trust in suppliers through information quality, and the establishment of trust pays more attention to consumers' perception of their kindness, integrity and ability. This paper proposes a food safety traceability technology based on blockchain, and combines experimental analysis to verify the effectiveness of this technology. After the food safety supervision information management system based on blockchain technology is successfully constructed and initially deployed, it changes the centralized data management mode under the traditional mode and achieves good application results. At the same time, it promotes regulatory scientific and technological innovation and improves regulatory efficiency.

Keywords—Blockchain; Food Safety; Traceability; Data association

I. INTRODUCTION

As businesses and global markets become efficient, competition in industries in the food sector is also shifting from "between businesses" to value chains, which need to provide added value to food. Although the modernization process has been advancing, due to the long supply chain, many production links, complex sources of ingredients and large turnover of personnel in the food field, many unfavorable factors have caused the society to lack confidence in food safety. Internet + blockchain technology is a new technology in the Internet age. It has the characteristics of decentralization and tamper-proof. Moreover, the whole process can realize anonymous machine autonomy and program audit. When applied in the food safety traceability system, it can solve the trust crisis and alleviate the waste of resources in a targeted manner [1].

Literature [2] has collected and sorted out the research and reports on food safety traceability based on Internet plus blockchain technology at home and abroad in recent years, sorted out the relevant concepts and basic principles, summarized the actual cases that have been implemented and the published research models, analyzed the current main advantages, disadvantages and development resistance of such systems, outlined the application of blockchain in the field of food safety traceability, as well as the existing

problems and challenges, provided important reference information for the follow-up improvement of food traceability system and food safety protection, and put forward suggestions and prospects for future food safety development, with a view to providing reference for the extensive application of Internet plus blockchain technology in the food field in the future

In the field of food safety, there is usually a massive amount of data information and heterogeneous data. In order to extract key information from it, knowledge graphs are widely used. The food safety knowledge graph includes various types, such as the food safety management knowledge graph, the food safety emergency response process knowledge graph, the food safety domain knowledge graph, the food hazard fingerprint graph, and the odor fingerprint graph[3]. Among them, the food safety management knowledge graph can provide regulatory authorities with detailed information on various links of food in the supply chain, and regulatory authorities can trace food issues through this type of knowledge graph. The knowledge graph of food safety emergency response process can assist relevant departments in handling similar food safety incidents, provide timely emergency measures, and effectively prevent and avoid the occurrence of similar incidents by crawling typical food safety case data from large food websites in recent years[4]. The knowledge graph in the field of food safety is the integration of multiple sources of food professional knowledge using knowledge graphs, which can more simply and intuitively popularize basic food safety knowledge to the public, and provide them with an additional guarantee for their physical health. Fingerprint mapping is a widely used knowledge graph in the food industry, which can be used to analyze various links in the food supply chain, detect whether food is adulterated, observe the mold growth of food during storage and transportation, and solve food safety issues such as the freshness of fruits and vegetables during harvesting, storage, and ripening[5].

In the food supply chain, building a knowledge graph oriented towards the field of food safety can help discover the connections between data at different nodes in the food supply chain, which is of great significance for solving problems such as data sharing, intelligent supervision of food safety, and traceability of the food supply chain[6]. However, there are still many problems with the circulation

of graph information throughout the entire lifecycle. Due to the fact that the food supply chain is a network structure that enables the circulation of food from producers to consumers, with numerous and highly dispersed participants in each link, there are problems such as poor flow of graph information, easy tampering of data, low traceability efficiency, and insufficient supervision. Therefore, the current food safety knowledge graph cannot achieve comprehensive and trustworthy management. In recent years, blockchain technology has developed extremely rapidly and is entering the 3.0 era of integration with various industries[7].

At present, blockchain technology has been applied in many fields such as education, healthcare, agriculture and food, supply chain management, etc. Among them, due to its characteristics of decentralization, immutability, and traceability, it has broad application prospects in the field of food safety, such as food supply chain traceability, food information security management, and food safety supervision[8]. According to research, combining the characteristics of blockchain technology can build a secure and reliable food safety traceability system, achieving the safety and traceability of food supply. In response to the problem of information asymmetry in various links of the food supply chain, blockchain technology can be used to ensure the transparency of transaction information, thereby expanding trust between enterprises and achieving food information security management [9]. In terms of food safety supervision, blockchain technology can break down information barriers between different departments, achieve information sharing between departments and between departments and supply chains, effectively utilize data resources throughout the entire industry chain, and greatly improve the efficiency of food safety supervision[10]. Due to the enormous advantages that blockchain can play in the field of food safety, a trustworthy management framework for food safety knowledge graph has been designed and proposed using blockchain technology. Technologies such as multi-mode storage mode, R-TBFT consensus mechanism, and customized smart contracts have been designed to ensure the reliable circulation and secure management of food safety knowledge graph information throughout the entire supply chain, providing new research ideas for the trustworthy management of food safety knowledge graph information throughout the entire chain [11].

II. WHOLE PROCESS MODEL OF FOOD SUPPLY CHAIN

The research on the whole process of food supply chain is very important for the research of traceability methods. This paper summarizes the whole process of food supply chain by systematically analyzing the production, processing, transportation and sales of food. The food supply chain process is mainly divided into five links, namely, raw material production link, raw material pretreatment, production and processing link, storage and transportation link and sales link.

First, the process information of the production link is

defined as fp :

$$fp = [G_1, G_2, G_3, Y_1, Y_2, Y_3, Y_4, Y_5, Y_6, Y_7] \quad (1)$$

Among them, G_1 is the name of the raw material production site, G_2 is the contact information of the raw material production site, G_3 is the address information of the raw material production site, Y_1 is the variety of raw materials, Y_2 is the origin of raw materials, Y_3 is the weight of raw materials, Y_4 is the production mode of raw materials, Y_5 is the production information, Y_6 is the safety information and Y_7 is the health status of raw materials.

As a key link in the food supply chain, raw material pretreatment is mainly the collection and management of raw materials. In this link, the activities designed to food safety issues can be summarized as follows: cleaning and disinfection operation of pretreatment equipment, raw material pretreatment operation process, raw material ingredient detection results, and information of relevant operators. Among them, cleaning and disinfection of pretreatment equipment is an operation that runs through the entire pretreatment process. This work should be carried out before the start of the pretreatment operation and after the end of the pretreatment operation, so as to ensure that the pretreatment process of food is aseptic. As a process for recording raw material collection, this paper defines raw material pretreatment process information as jp :

$$jp = [I_1, I_2, I_3, I_4, I_5, I_6, I_7, I_8] \quad (2)$$

Among them, I_1 is the sanitary status of the pretreatment equipment, I_2 is the job number of the pretreatment personnel, I_3 is the protein content of the raw materials, I_4 is the fat content of the raw materials, I_5 is the toxin content of the raw materials, I_6 is the heavy metal content of the raw materials, I_7 is the abnormal element content of the raw materials, and I_8 is the type and content of bacterial colonies.

Production and processing link is the most complicated and critical link in food supply chain. Therefore, this paper collects all links of production and processing based on component indexes. In this way, it can ensure that the node analysis of food safety problems can be realized in an all-round way, and provide a more complete information environment for the traceability and location of food safety. In the actual production and processing regulations, driven by interests, some businesses will have illegal operations. By adding illegal raw materials in the food processing process, the smell or taste of products can be improved and the production cost can be reduced at the same time.

However, such practices often have a negative impact on consumers. Therefore, in order to realize the accurate traceability of the processing link, this paper defines the information mp of the production and processing link as follows:

$$mp = [A_1, A_2, A_3, A_4, A_5, A_6, A_7, A_8] \quad (3)$$

Among them, A_1 is the address of the processing plant, A_2 is the quality inspection result of the raw materials, A_3 is the aseptic operation mode of the raw materials, A_4 is the ingredients and specific content of the food, A_5 is the time of the processing process, A_6 is the inspection result of the finished food, A_7 is the packaging method of the product, and A_8 is the time of the packaging process.

After the food processing is completed, the products with compound inspection standards enter the storage and distribution links. The main factors that will affect food safety in this link are food storage temperature and logistics and distribution time. Therefore, in order to facilitate the tracing of the factors that affect food quality and safety in this link, this paper sets the information tp of storage and transportation link as follows:

$$tp = [S_1, S_2, S_3, S_4, S_5, S_6, S_7] \quad (4)$$

Among them, S_1 is the start time of commodity storage, that is, the time when it enters the storage warehouse and the time when the commodity leaves the warehouse, S_2 is the temperature of the environment in which the commodity is stored, S_4 is the temperature of the environment in which the commodity is located during transportation, S_5 is the transit node in transportation, S_6 is the destination and S_7 is the corresponding transportation number.

As the last step of the food supply chain, the sales link is very important for the information traceability of food safety issues. In this link, the factors affecting food safety mainly include the sales environment. Therefore, the traceability information collection of this link in this paper mainly focuses on the specific sales of food. this paper defines the sales link information sp as:

$$sp = [B_1, B_2, B_3, B_4, B_5, B_6] \quad (5)$$

Among them, B_1 is the name of the seller, B_2 is the sales address, B_3 is the contact information, B_4 is the sales time, B_5 is the quantity and B_6 is the sales staff information.

To sum up, the whole process traceability information model can be expressed as:

$$e = [fp.inf \cup jp.inf \cup mp.inf \cup tp.inf \cup sp.inf] \quad (6)$$

On this basis, around these traceability information, the traceability and location calculation of food safety problems are realized.

III. SYSTEM CONSTRUCTION

In terms of application system construction, all social participants run an information platform according to a set of systems, a standard and a data center. Through social co-governance, a traceability collaboration mechanism is constructed, and the application architecture platform design is shown in Figure 1.

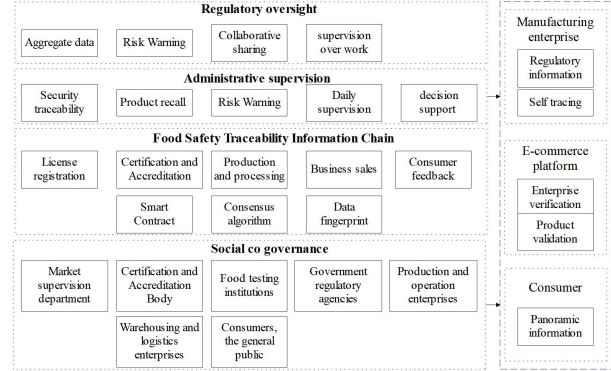


Figure 1. Application architecture platform

The food safety traceability collaboration platform based on blockchain connects with information systems such as production enterprises, suppliers, sellers, logistics enterprises and technical institutions to obtain true and credible traceability information, provide necessary inquiry services to consumers and the public, and provide data support for the supervision of market supervision departments. The specific technical architecture is shown in Figure 2.

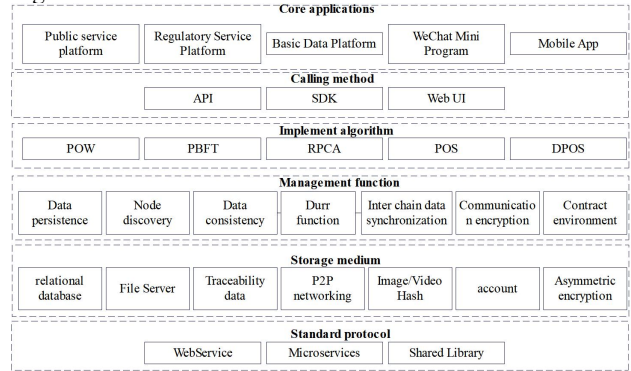


Figure 2. Technical architecture platform

At the level of planning the blockchain service platform, it is necessary to require the system to have good scalability for the diversity and complexity of trusted transaction scenarios, follow the modular design principle, and provide the minimum external expansion boundary while ensuring

the stability of the core logic of the system to achieve high cohesion and low coupling of the system. At the same time, the trustworthiness of blockchain needs to follow security principles, data auditability principles, and meet standards and compliance requirements for different regional scenarios in system design and implementation, so as to ensure that information processing can meet confidentiality, integrity, and controllability, availability and non-repudiation requirements. The schematic diagram of the blockchain architecture is shown in Figure 3.

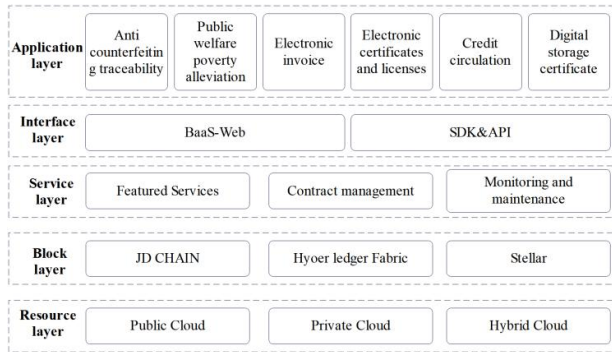


Figure 3. Schematic diagram of blockchain architecture

This paper combines blockchain technology to design blockchain database management, which makes all operation information of the database complete and reliable records, and avoids the possibility of data tampering. The blockchain-based database management system combines the underlying database technology with blockchain technology, uses the traditional database to store and record the original data, and writes the user's operation log of the database into the blockchain for distributed storage. In this process, the user can autonomously use the Pc or mobile client to store and query the required data. The specific operation is shown in Figure 5.

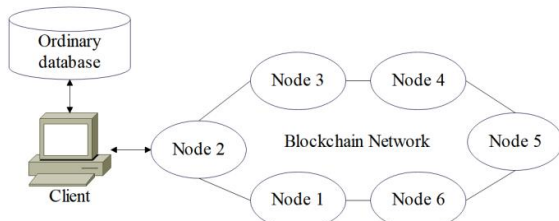


Figure 4. Blockchain database flow chart

The regulatory authority can monitor the authenticity of the information in the supply chain process, and if there is a problem in a certain link, it can immediately track the specific responsible person through this method, as shown in Figure 5.

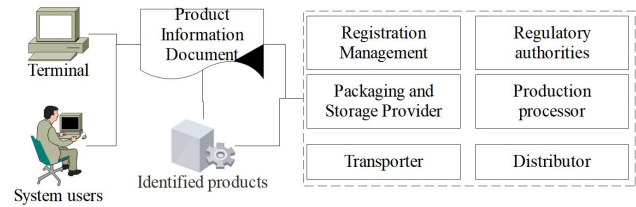


Figure 5. Architecture diagram of product traceability network based on blockchain

After the user is successfully registered, each user will generate a public key and a private key based on the IP and wallet address, and at the same time, under the insurance measures of identity information verification, the data security information will be matched best. Among them, the public key is connected to the data transmission chain and logged in to the network in the form of public information. The whole system participants can directly query the data in the public key, and at the same time, they can also verify the authenticity of the information. However, the private key is saved by the user himself, which is the key to logging in to the food system to obtain data and reversely verifying the identity of consumers or regulators. With a well-debugged smart contract drawn up in advance, the generation and transmission of data throughout the production and sales process does not require much manual intervention. At the same time, the technical code stored in independent isolation conditions makes it impossible for anyone to modify the operating rules of the traceability mechanism. Therefore, the blockchain-based step-by-step chicken traceability method can ensure the authenticity and reliability of traceability information, and the data in the system will not be arbitrarily changed by illegal nodes.

IV. TESTING

10 pieces of food of a certain brand with food safety problems are set as test objects, 4 pieces of safety problems are set in the transportation link, numbered 1-4, 5 pieces of safety problems are set in the processing link, numbered 5-9, and 1 piece of safety problem is set in the consumer link, numbered 19. Moreover, 10 products with safety problems are traced and located by the methods of reference [6], reference [7] and this paper respectively, and the results are compared

(1) Traceability and location results of safety issues. Under the above test environment, the positioning results of the three methods are compared, as shown in Table I.

TABLE I. TRACEABILITY AND LOCATION RESULTS OF DIFFERENT METHODS

	Method of reference [6]	Method of reference [7]	Methods of this paper
Transportation link	4	4	4
Processing link	4	4	5
Consumer link	0	1	1
Other	1	0	0

In the design method in this paper, not only the 100% accuracy of safety problem location is guaranteed, but also the problem location results show 100% integrity, which shows that the designed method has high practical application value in traceability and location of food safety problems. The main reason for this is that this paper realizes the model coding of food information, and accurately finds the correct problem source in the coding through the blockchain consensus method.

(2) Time consumption of traceability and location of security issues. On the basis of the above, the time-consuming situation of the three methods to obtain the positioning results is counted, and the results are shown in Table II.

TABLE II. TIME CONSUMPTION FOR COMPLETING TRACEABILITY AND LOCATION (UNIT: S)

Serial Number	Method of Reference [6]	Method of Reference [7]	Methods of this Paper
1	2.28	2.39	1.84
2	2.27	2.26	1.95
3	2.31	2.01	1.71
4	2.26	2.29	1.99
5	2.23	2.35	1.84
6	2.27	2.01	1.99
7	2.36	2.13	1.71
8	2.27	2.33	1.83
9	2.38	2.32	1.72
10	2.27	2.27	1.88
Mean value	2.29	2.24	1.85

By comparing the average time consumption of the three methods, it can be seen that the time consumption of the method of this paper can basically meet the actual needs. This shows that the proposed method does not cause additional time overhead on the premise of improving the positioning accuracy, and has good practical application value

V. CONCLUSION

Strengthening the research on food safety is an important foundation for the safe, stable and orderly operation of the market environment, and at the same time, it is of extremely important value and significance to the healthy development of human body. As a fundamental basis to solve problems, food safety traceability is an extremely important application means after food safety problems occur. Therefore, accurate and reliable positioning results are an important guarantee for the control of food safety problems. This paper proposes a research on the traceability and location method of food safety problems based on blockchain consensus, which achieves the purpose of improving the location accuracy and reliability, and has certain practical application value for solving actual food safety problems. Therefore, through this research, it is

hoped to provide valuable reference for the research of traceability and location, and also provide some help for food safety guarantee.

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